

Deep Energy-Efficient Optimization Network for URLLC Over Cell-Free Massive MIMO

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Abstract—To achieve ultrareliable and low-latency communication (URLLC) and support high density of wireless connections simultaneously, the sixth-generation Industrial Internet of Things (6G-IIoT) necessitates an expansion of antenna arrays and broader bandwidths, which suffers from high energy consumption. To address this issue, this article investigates a cell-free massive multiple-input-multiple-output (CF-mMIMO) system and designs an iterative search-based two-stage energy efficiency (EE) optimization algorithm for the uplink communication of the system. The first stage prioritizes reliability to ensure that all users meet the URLLC requirements regarding latency and reliability. The second stage maximizes the EE under URLLC-satisfied conditions. Considering that iterative search algorithms incur a high computational overhead and variable number of iterations, we further propose a convolutional neural network architecture (RACNN) to approximate an optimal resource allocation strategy and to realize the real-time and stable output. This structure extracts deep correlations among users from the global channel features. It takes strategies of multitask learning and weight loss adaptation to improve the model's convergence speed. Furthermore, we employ deep transfer learning to adjust RACNN parameters to accommodate the potential of dynamic communication scenarios, thereby reducing the demand for training samples and training time overhead. Finally, the efficacy of the proposed algorithm, RACNN, and deep transfer learning is validated through experimental simulations.

Index Terms—Cell-free massive multiple-input-multiple-output (CF-mMIMO), deep learning, energy efficiency (EE), ultrareliable and low-latency communication (URLLC).

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I. INTRODUCTION

IN THE context of the sixth-generation Industrial Internet of Things (6G-IIoT), the smart factory concept has attracted considerable attention owing to its intelligent production environment and efficient resource management [1], [2], [3]. In this scenario, numerous terminals intelligently access the network in real-time interaction, which poses significant challenges to the latency and reliability of wireless communication [4]. Massive ultrareliable and low-latency communication (mURLLC), which takes into account the Quality of Service (QoS) constraints on end-to-end latency and error rate bounds, has been widely adopted in IIoT to ensure high reliability, low latency, and large-scale connectivity [5], [6], [7]. However, the energy consumption issue is not emphasized in the mURLLC protocol, but it is essential for 6G-IIoT scenarios, continuously determining whether the whole system works appropriately or not [8]. Besides, considering that most terminals in ultrareliable and low-latency communication (URLLC) transmit signals with short packets, the traditional Shannon information theory does not apply to finite blocklength (FBL) transmission [9]. Therefore, accurately characterizing the relationship between latency, reliability, and energy efficiency (EE) in mURLLC-adopted IIoT application scenarios and designing a targeted optimization algorithm for energy consumption is an interesting and worthwhile research [10].

Zander [11] and Foschini and Miljanic [12] investigated the feasibility and specific schemes for performance optimization through iterative algorithms. However, implementing these algorithms in practical networks encounters two primary challenges. First, optimization problems are generally nonconvex. These approaches seek to identify the optimal strategy by traversing the solution space, which leads to considerable computational overhead. Even with specific well-established methods for solving nonconvex optimization problems, such as the interior-point method [13], the complexity of the computation is too high to achieve stable and real-time communication. Second, the system must update its resource allocation via extensive simulations and experiments to handle the changing channel conditions, which also causes a lengthy process latency [14].

Recently, deep neural networks (DNNs) [15] have been employed extensively in IIoT to find the optimal resource allocation strategy [16], [17], [18], [19]. The basic idea is to utilize a DNN to approximate the mapping relationship between the system state and allocation strategy. Initially, numerous labeled

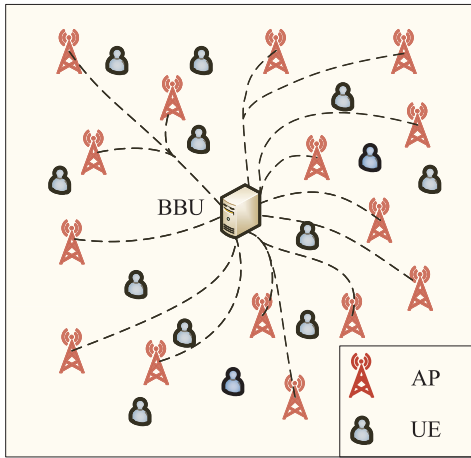


Fig. 1. Framework of CF-mMIMO.

samples will be generated through a known resource allocation algorithm. Subsequently, a DNN is trained adaptively through supervised learning. After the training stage, the network parameters are fixed, and the DNN can generate the optimal resource allocation from any input channel state. According to the universal approximation theory [20], if the number of neurons in a network is sufficiently large, the approximation errors will converge to zero indefinitely. However, owing to the massive antennas and user equipment (UE) in IIoT and resource allocations for different UEs that are often interrelated, these present a significant challenge to the DNN's generalization abilities. Additionally, the network is trained offline. When the wireless network is nonstationary, some hidden variables not included in the training data may change, leading to the model violating its robustness range [21].

A. Related Work

1) *Cell-Free Massive MIMO*: Cell-free massive multiple-input-multiple-output (CF-mMIMO) represents a key technology for 6G [22], [23], [24]. As shown in Fig. 1, the core concept is to decentralize the deployment of a large number of access points (APs) within a specific area, which are connected to a baseband unit (BBU). Elhoushy et al. [25] provided a comprehensive overview of CF-mMIMO systems, including general system modeling, detailed system operation, implementation limitations. Furthermore, they explored opportunities to integrate the system with emerging technologies, raised open questions, and outlined future research directions. Freitas et al. [26] developed a framework for selecting scalable APs based on a competitive mechanism. They proposed three strategies for fine-tuning the access point clustering to reduce the number of UEs per access point without affecting the spectral efficiency (SE) or improving the EE. In a CF-mMIMO system, a large amount of APs distributed deployment significantly reduces the distance between UEs and APs, provides robust anti-fading capabilities and enhances UE performance compared to traditional mMIMO systems. Therefore, it is more suitable for supporting URLLC services [24], [25], [26].

URLLC is involved in many critical IIoT scenarios, including tele-surgery, industrial robotics, and autonomous driving [6]. These scenarios demand rigorous latency and reliability, and even minor delays or packet loss causes serious consequences [25]. Currently, the research for URLLC in CF-mMIMO is relatively mature. Zhang et al. [23] considered downlink multidevice CF-mMIMO with a hard deadline and utilized two instantaneous evaluation metrics to characterize the latency and reliability of the communication. Huang et al. [27] investigated the SE and EE performance of CF-mMIMO-assisted URLLC for short packet communication. Initially, they derived the system's error probability (EP) and reachability rate, providing a more accurate characterization of its performance. Subsequently, they analyzed the EE based on a typical power consumption model. Bucci et al. [28] investigated the potential of CF-mMIMO systems to support both URLLC and eMBB services through network slicing and the utilization of spatial diversity, establishing a theoretical framework to evaluate the comprehensive performance of the system. However, the substantial access of UEs and multi-antenna APs in CF-mMIMO complicates resource management, with additional complexity heightened by URLLC constraints. Therefore, it is necessary to design rational resource allocation methods to improve communication performance based on guaranteed URLLC requirements [23], [27], [28].

2) *Traditional Resource Allocation in Cell-Free Massive MIMO*: Many studies have focused on designing a flexible and efficient resource allocation scheme to enhance communication performance [29]. For example, Nasir et al. [30] developed a path-following algorithm for URLLC downlink. This algorithm can output a series of improved feasible points to optimize the transmission rate and system EE of URLLC. Fang et al. [31] derived a closed-form expression for the downlink achievable rate of normalized conjugate beamforming (NCB) in CF-mMIMO systems to maximize the minimum user rate. Subsequently, they formulated a max-min power optimization model with power constraints at each AP. This model is addressed using a path-following algorithm, which approximates the objective function as a logarithmic function, converts polynomial constraints to monomial constraints, and finally transforms the original problem into a series of geometric programming problems to be solved iteratively. Peng et al. [32] employed CF-mMIMO in a smart factory under URLLC. They first establish lower bounds on achievable rates for FBL uplinks and transform the nonconvex weighting and rate optimization problem. Then, an iterative algorithm is developed to jointly optimize pilot frequency and payload power allocation. However, these resource optimization algorithms tend to find a solution by iteration, which leads to a significant computational overhead [29], [30], [31], [32].

3) *Deep Learning for Resource Allocation in Cell-Free Massive MIMO*: In recent years, many scholars have attempted to learn the optimal resource optimization strategy for CF-mMIMO systems using deep learning [33]. Huang et al. [34] proposed a deep reinforcement learning (DRL) method for optimizing the power allocation strategy for public and private flows. The method can obtain higher

EE than space division multiple access DRL (SDMA-DRL) and traditional power allocation algorithms. Yan et al. [35] investigated power control from the uplink and downlink perspectives. They developed a neural network-assisted power control algorithm, drastically reducing the computation time and significantly improving the system SE. Jin et al. [36] proposed a channel estimation framework based on denoising convolutional neural network (FFDNet), which is suitable for various levels of signal-to-noise ratios compared to traditional deep learning-based channel estimation methods that take noise level maps as inputs. Salaün and Yang [37] designed a deep learning approach to find suboptimal power control strategies for CF-mMIMO systems. Specifically, a convolutional neural network (CNN) is proposed to process a channel matrix with large-scale fading coefficients as input, predicting the total transmit power for each AP. Subsequently, the authors utilize a low-complexity convex optimization algorithm to compute the downlink power control strategy for each user. Khan et al. [38] proposed a DNN framework to execute power controlling and pilot allocating. Furthermore, the coefficients of the same users are compressed as the network inputs for reducing computing overhead. Furthermore, It is noteworthy that the distribution of wireless channels in the network may change over time. These changes are called hidden variables and are not included in the training datasets, posing potential challenges in deep learning [35], [36], [37], [38].

B. Contribution

From existing studies, we can see that EE optimization has been extensively explored in CF-mMIMO scenarios, with a growing body of related research. However, the joint optimization of EE, latency, and reliability has been rarely discussed. Furthermore, traditional iterative optimization algorithms consume considerable computational resources, which is not conducive to real-time communication in IIoT.

This article addresses the EE maximization problem for the uplink of CF-mMIMO systems under the constraint of URLLC demand. Inspired by the abovementioned considerations, we devise a power allocation algorithm to maximize EE. Then, we develop a DNN to approximate this strategy and enable stable real-time output of the power allocation strategy determination. Finally, we incorporate transfer learning to enhance the model's efficiency and scalability and accommodate changing communication scenarios. Overall, the main contributions of our article are as follows.

- 1) We employ FBL information theory [39] to accurately analyze and model EE, reliability, and end-to-end latency in CF-mMIMO systems. Subsequently, a two-stage energy-efficiency optimization algorithm is designed to maximize the EE by adjusting the power allocation while achieving URLLC. The first stage ensures that all UEs meet the latency and reliability requirements for URLLC, and the second stage builds on this foundation by optimizing EE. The algorithm enables the optimal power allocation strategy and maximizes the number of URLLC users in the system.

- 2) We propose a resource allocation convolutional neural network (RACNN) to approximate the proposed power allocation strategy efficiently. RACNN eliminates extensive iterative searches, thus achieving stable real-time power output. Initially, the global semantic features of all UEs are fully extracted by convolutional layers. These features are then mapped to the known power allocation by fully connected layers. Furthermore, we introduce a multitask learning architecture to accelerate model convergence by learning the relationship between optimal power allocation and maximum EE. Finally, we employ a dynamic weights method to assign weight ratios to different tasks, thereby improving inference speed.
- 3) We utilize deep transfer learning [40] to fine-tune pre-trained DNNs in a nonstationary wireless environment. This approach involves reusing the initial layers of a pretrained DNN and training only the final layers, which can be used to train a high-quality model with fewer samples and less time than retraining a neural network.

C. Organization

The remainder of this article is organized as follows. In Section II, we introduce a CF-mMIMO system architecture, detail the channel estimation and multiuser detection process, and analyze the relationship between reliability, latency, and EE in short packet transmission. Section III proposes a model and optimization algorithm for maximizing EE in URLLC. A DNN for approximating optimal resource allocation and a deep transfer learning method is designed in Section IV. The simulation validation of the proposed method is presented in Section V. The article concludes in Section VI.

Notations: $(\cdot)^H$ and $(\cdot)^T$ denote the conjugate transpose and transpose, respectively. $\|\cdot\|$ is the Euclidean norm, $\mathcal{N}(0, 1)$ and $\mathcal{CN}(0, 1)$ are Gaussian distribution and complex Gaussian distribution with mean zero and unit variance, respectively. $\mathbb{E}(\cdot)$ denotes the expectation operator. $\mathbb{C}^{K \times M}$ and $\mathbb{R}^{K \times M}$ represent real and complex matrix with K rows and M columns, respectively.

II. SYSTEM MODEL AND PROBLEM FORMULATION

Consider a CF-mMIMO system deployed in a smart factory. The system comprises M APs serving K UEs across the coverage area. As shown in Fig. 2, each AP is equipped with L antennas, while each UE is equipped with a single antenna. Given that all UEs share the same time-frequency resources, it is necessary to allocate appropriate time-frequency block resources to each UE while effectively controlling the transmit power of each node. Assuming that the uplink from the UE to the AP is modeled as a block fading channel model, the time-varying broadband channel is divided into time-frequency coherent blocks [41]. Furthermore, we consider that the data received from all APs are aggregated at the BBU via error-free backhaul links. The BBU then makes global decisions regarding allocating time-frequency resources and transmitting power control.

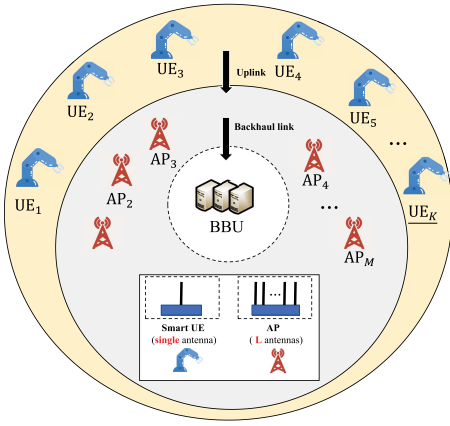


Fig. 2. CF-mMIMO in smart factory.

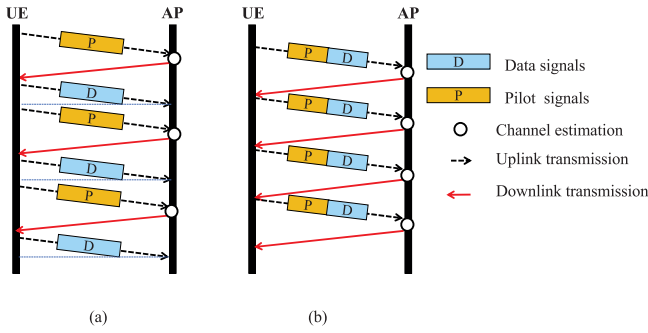


Fig. 3. Transmission method of pilot and data signals. (a) Separate transmission of pilots and data signals. (b) Simultaneous transmission of pilots and data signals.

A. Uplink Channel Estimation

Accurate channel estimation plays a crucial role in adapting to channel conditions, optimizing resource allocation, and scheduling, which enhance the system's throughput, reliability, and efficiency [42]. Conventional channel estimation methods utilize pilots to acquire channel state information (CSI), as shown in Fig. 3(a). Specifically, the UE transmits a pilot sequence to the AP at a specified time interval. However, considering that terminals in IIoT application scenarios are exposed to varied factors, such as mobile devices, obstacles, and electromagnetic interference, the channel tends to exhibit fast-fading characteristics, posing challenges to achieving accurate and real-time channel estimation. In order to satisfy the requirements of URLLC for short packet services in IIoT, it is necessary to obtain real-time CSI [43]. The pilot sequence is encapsulated with the transmitted data into a short packet to acquire real-time pilot, as shown in Fig. 3(b). This method sacrifices resource efficiency for improved data transmission reliability. We consider that each UE transmits a short packet containing D bits of data through N channel uses (CUs), where $N = Bt_D$, B is the CU bandwidth, t_D is the end-to-end CU latency, and all CUs take the same transmit power [44]. The channel coefficient between the k th UE and the m th AP is with Rician distribution [23], formulated as

$$\mathbf{g}_{m,k} = \sqrt{\frac{\rho\beta_{m,k}}{\rho+1}}\mathbf{h}_{m,k}^{(L)} + \sqrt{\frac{\beta_{m,k}}{\rho+1}}\mathbf{h}_{m,k}^{(N)} \quad (1)$$

where ρ represents the Rician factor, $\beta_{m,k}$ denotes large-scale factor (LSF) contains path loss and shadows effects between the k th UE and the m th AP, $\mathbf{h}_{m,k}^{(L)} = [e^{j\theta_{m,k,1}}, \dots, e^{j\theta_{m,k,L}}]^T \in \mathbb{C}^{L \times 1}$ is the random variable vector of Line-of-Sight (LoS), $j = \sqrt{-1}$, $\theta_{m,k,l} (l \in [1, \dots, L])$ is the random variable with uniformly distributed between $[-\pi, \pi)$, $\mathbf{h}_{m,k}^{(N)} \in \mathbb{C}^{L \times 1}$ is the random variable of nonline-of sight (NLoS) with independent-identically distributed (i.i.d.) $\mathcal{CN}(0, 1)$.

The received signal of the m th AP $\mathbf{Y}_m \in \mathbb{C}^{L \times N}$ is expressed as

$$\mathbf{Y}_m = \sum_{k=1}^K \mathbf{g}_{m,k} \sqrt{p_k} [\mathbf{x}_{(p),k}, \mathbf{x}_{(s),k}] + \mathbf{W}_m \quad (2)$$

where p_k is the UE transmit power, $\mathbf{x}_{(p),k} \in \mathbb{C}^{1 \times \tau_{(p)}}$ and $\mathbf{x}_{(s),k} \in \mathbb{C}^{1 \times \tau_{(s)}}$ represent the pilot signal and data signal with blocklength $\tau_{(p)}$ and $\tau_{(s)} = N - \tau_{(p)}$ (in CUs), respectively, $\mathbf{W}_m = [\mathbf{W}_{(p),m}, \mathbf{W}_{(s),m}] \in \mathbb{C}^{L \times N}$ expresses as the additive gaussian white noise (AWGN) with i.i.d. $\mathcal{CN}(0, 1)$.

To avoid the impact of pilot pollution, different UEs are assigned mutually orthogonal pilot signal with blocklength, i.e., $\mathbf{x}_{(p),k} \mathbf{x}_{(p),k}^H = \tau_{(p)}$, $\mathbf{x}_{(p),k} \mathbf{x}_{(p),k'}^H = 0$ ($k' \neq k$), and $\mathbb{E}(|\mathbf{x}_{(p),k}|^2) = 1$. The data signal $\mathbf{x}_{(s),k}$ is unrelated to each other, i.e., $\mathbb{E}(|\mathbf{x}_{(s),k}|^2) = 1$ and $\mathbb{E}([\mathbf{x}_{(s),k}][\mathbf{x}_{(s),l}]^H) = 0$, where $i \neq k$ or $j \neq l$.

After extracting the pilot sequence from the received signal, the channel is estimated at the APs. Typical estimation approaches include least squares (LS), minimum mean square error (MMSE), and interpolation methods [44]. This article selects the low-complexity LS algorithm for channel estimation to minimize processing latency. Define $\mathbf{W}_{(p),m} \in \mathbb{C}^{L \times \tau_{(p)}}$ as the pilot noise, the estimated channel between the k th UE and m th AP $\hat{\mathbf{g}}_{m,k}$ is expressed by

$$\begin{aligned} \hat{\mathbf{g}}_{m,k} &= \frac{(\mathbf{g}_{m,k} \sqrt{p_k} \mathbf{x}_{(p),k} + \mathbf{W}_{(p),m}) \mathbf{x}_{(p),k}^H \sqrt{p_k}^{-1}}{\tau_{(p)}} \\ &= \mathbf{g}_{m,k} + \frac{\mathbf{W}_{(p),m} \mathbf{x}_{(p),k}^H \sqrt{p_k}^{-1}}{\tau_{(p)}}. \end{aligned} \quad (3)$$

Following the channel estimation, the data signal matrix of the m th AP $\mathbf{Y}_{(s),m} \in \mathbb{C}^{L \times \tau_{(s)}}$ can be expressed as

$$\mathbf{Y}_{(s),m} = \sum_{k=1}^K (\hat{\mathbf{g}}_{m,k} - \mathbf{n}_{m,k}) \sqrt{p_k} \mathbf{x}_{(s),k} + \mathbf{W}_{(s),m} \quad (4)$$

where $\mathbf{n}_{m,k} = [\mathbf{W}_{(p),m} \mathbf{x}_{(p),k}^H \sqrt{p_k}^{-1} / \tau_{(p)}]$ is the estimation error vector between the k th UE and m th AP, with i.i.d. $\mathcal{CN}(0, [1/\tau_{(p)} p_k])$. $\mathbf{W}_{(s),m} \in \mathbb{C}^{L \times \tau_{(s)}}$ denotes the data noise.

B. Joint Multiuser Detection

Considering that all UEs share the same time-frequency resources, interuser interference will significantly impact the accurate decoding of the signals at the receiving end [23]. Therefore, it is crucial to detect and separate each UE's signal from the superimposed signal quickly and accurately. This article proposes a two-stage multiuser detection (MUD) algorithm. In the first stage, each AP combines the received

signals from multiple receive antennas using maximum ratio combining (MRC) [25] method, with the weights set to the estimated channels $\hat{\mathbf{g}}_{m,k}^H$.

Subsequently, all APs transmit the output signals to the BBU for joint detection. The BBU then weights the signals of each AP to separate the desired signals [45]. Assuming that the large-scale fading coefficients can be predicted at the receivers [44]. The weight setting depends on the normalized large-scale fading coefficient to prevent excessive channel quality gaps. That is, the weight of the k th UE's signal in the m th AP is set to be $\alpha_{m,k} = \beta_{m,k} / \sum_{i=1}^K \beta_{m,i}$ [44]. Compared to traditional MUD methods, the proposed detection approach effectively improves the signal-to-interference-plus-noise ratio (SINR) for target UEs while distributing the computational burden across the APs, enhancing detection efficiency. The detected signal of the k th UE in the BBU $\mathbf{y}_{(s),k} \in \mathbb{C}^{1 \times \tau_{(s)}}$ is given by

$$\begin{aligned} \mathbf{y}_{(s),k} &= \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \mathbf{Y}_{(s),m} \\ &= DS_k \cdot \mathbf{x}_{(s),k} + \sum_{i=1, i \neq k}^K UI_{k,i} \cdot \mathbf{x}_{(s),i} \\ &\quad - \sum_{i=1}^K EM_{k,i} \cdot \mathbf{x}_{(s),i} + AW_k \end{aligned} \quad (5)$$

where

$$DS_k \triangleq \sqrt{p_k} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,k} \quad (6)$$

$$UI_{k,i} \triangleq \sqrt{p_i} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,i} \quad (7)$$

$$EM_{k,i} \triangleq \sqrt{p_i} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \mathbf{n}_{m,k} \quad (8)$$

$$AW_k \triangleq \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \mathbf{W}_{(s),m} \quad (9)$$

denote the desired signal, interuser interference, channel estimation error, and AWGN of the k th UE, respectively.

Theorem 1: Assuming the statistics of the channel and estimation errors are known at the terminals, along with the premise of a set of deterministic large-scale coefficients. The sum of terms in (7), (8), and (9) is treated as effective noise [22]. For simplicity, let $\hat{\xi}_{m,k} = \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}|^2\}$, and $\hat{\mu}_{m,i,k} = \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,i}|^2\} (i \neq k)$, the SINR for the k th UE can be calculated by (10), shown at the bottom of the next page [22], [44].

Proof: Please refer to Appendix. ■

C. Performance Analysis

In the CF-mMIMO system mentioned in this article, all UEs must meet URLLC requirements on latency and reliability to ensure real-time information exchange. Moreover, a large number of UEs access the network, and significant energy consumption is observed. Therefore, optimizing the EE

under URLLC constraints becomes crucial. First, we analyze latency components and develop closed-form expressions for reliability using FBL information theory. Subsequently, the computational expression of EE is presented. Finally, we analyze the relationship between different performance metrics, which will be used for subsequent performance optimization.

We define the end-to-end latency of the uplink communication process as t_D [43]. That is, the period begins when the signals are transmitted and continues until the signals are decoded. During this interval, t_D consists of transmission latency t_T , propagation latency t_P , retransmission latency t_R , and processing latency t_S , which is expressed by

$$t_D = t_T + t_P + t_R + t_S. \quad (11)$$

In the considered scenario, multiple APs are deployed and decentralized within an area, and the transmitter and receiver positions are fixed. Thus, the value of t_P is a tiny constant. Meanwhile, t_S is minimal due to the APs simultaneously receiving and processing the signals in parallel. Furthermore, URLLC has stringent reliability requirements. t_R is not considered. Therefore, t_D can be approximately as t_T . The latency boundary for the UE under URLLC requirements is as follows:

$$t_D \leq t_{\max}. \quad (12)$$

As previously stated, a UE encapsulates the pilot sequence with the transmitted data into a short packet, and all UEs share the same time-frequency resources. The transmission latency t_T is decomposed to

$$t_T = t_{PT} + t_{ST} \quad (13)$$

where t_{PT} represents the pilot latency, t_{ST} is the data latency.

In addition to the latency, reliability is a significant metric for URLLC. The interruption probability and EP are generally used for evaluating the reliability of a communication system [46]. However, considering the ultrahigh reliability of URLLC, the probability of system interruption is extremely low. Consequently, this article evaluates the system's reliability in terms of the decoding EP. According to the FBL information theory [39], the relationship between data rate R_k and EP $\mathcal{E}_{(s),k}$ of the k th UE is modeled as

$$R_k \approx C(\gamma_k) - \frac{\sqrt{V(\gamma_k)}}{\tau_{(s)}} Q^{-1}(\mathcal{E}_{(s),k}) \quad (14)$$

where $C(\gamma_k) = \log_2(1 + \gamma_k)$ represents the Shannon capacity, $V(\gamma_k) = 1 - (1 + \gamma_k)^{-2} \log_2^2 e$ denotes the channel dispersion parameter. The probability that a Gaussian random variable x with mean 0 and variance 1 is greater than w is evaluated by the function $Q(w)$, which is defined as $Q(w) = (1/\sqrt{2\pi}) \int_w^\infty e^{-t^2/2} dt$.

Recall that each UE transmits D bits information, includes the data blocklength of $\tau_{(s)}$, the data rate is expressed by

$$R_k = \frac{D}{\tau_{(s)}}. \quad (15)$$

Therefore, the EP $\mathcal{E}_{(s),k}$ is further expressed as

$$\mathcal{E}_{(s),k} = Q\left(\left(C(\gamma_k) - \frac{D}{\tau_{(s)}}\right) \sqrt{\frac{\tau_{(s)}}{V(\gamma_k)}}\right). \quad (16)$$

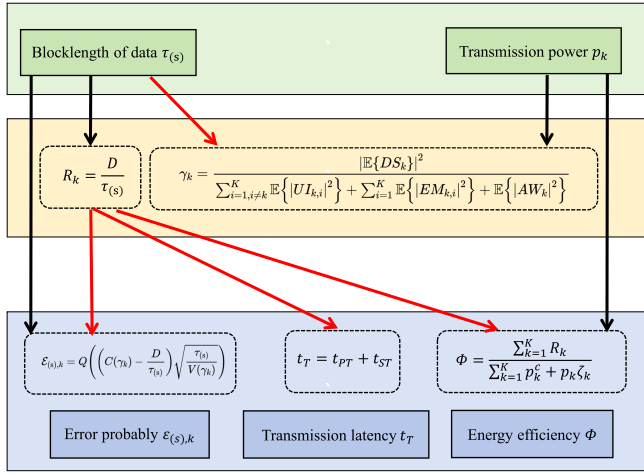


Fig. 4. Relationship between latency, reliability, and EE.

The EP boundary under URLLC constraints is

$$\mathcal{E}_{(s),k} \leq \mathcal{E}_{\max}. \quad (17)$$

In a CF-mMIMO-based URLLC system, EE is defined as the ratio of the achievable data rate to the total energy consumption, which is expressed as (in bits/J) [35]

$$\Phi = \frac{\sum_{k=1}^K R_k}{\sum_{k=1}^K p_k^c + p_k \zeta_k} \quad (18)$$

where p_k^c is the power consumption of the k th user transmitter circuit and ζ_k represents the reciprocal of the drain efficiency power amplifier.

Further analysis of the relationship between latency, reliability, and EE is depicted in Fig. 4. The primary factors influencing these metrics are the blocklength of data $\tau_{(s)}$ and transmit power p_k . The change in $\tau_{(s)}$ directly affects SINR, data rate, and EP and further impacts latency and EE. Furthermore, the change in p_k directly affects SINR and EE and indirectly impacts EP. Therefore, it can be demonstrated that latency, reliability, and EE constraints are complex coupled, which poses a significant challenge to optimize EE while meeting URLLC in our considered system.

III. ENERGY EFFICIENCY OPTIMIZATION

As mentioned, the optimization problem may be modeled by the blocklength of data $\tau_{(s)}$ and transmit power p_k . However, jointly optimizing both $\tau_{(s)}$ and p_k is a nonconvex issue, and meanwhile, the optimization of $\tau_{(s)}$ is a complex process, leading to high computational complexity. To address these

challenges, we fix the blocklength of data $\tau_{(s)}$ and optimize EE by adjusting p_k . The optimization model is designed as

$$\begin{aligned} \max_{p_k} \quad & \Phi \\ \text{s.t.} \quad & p_k \leq p_{\max} \\ & 1 \leq \tau_{(s)} \leq \tau_{\max} \\ & R_{\min} \leq R_k \\ & \mathcal{E}_{(s),k} \leq \mathcal{E}_{\max} \\ & t_D \leq t_{\max} \end{aligned} \quad (19)$$

where $p_{\max}$, $\mathcal{E}_{\max}$, and $t_{\max}$ represent the maximum transmit power of UEs, EP, and latency boundary, respectively, $\tau_{\max} = N - K$ is imposed to preserve pilot orthogonality, $R_{\min} = D/\tau_{\max}$ denotes the minimum target rate constraint.

According to (10), shown at the bottom of the page and (16), the trend of transmit power p_k is found to be positively correlated with the SINR γ_k and negatively correlated with the EP $\mathcal{E}_{(s),k}$. Therefore, there exists a lower bound, p_k^* , such that $\mathcal{E}_{(s),k}$ reaches the boundary $\mathcal{E}_{\max}$. However, a reduction of p_k results a decrease interuser interference, thereby increasing for γ_i ($i \leq K$ and $i \neq k$), which will further lower p_i^* . Recall that $\tau_{(s)}$ is fixed to a value that satisfies all the constraints. (19) can be simplified as

$$\begin{aligned} \max_{p_k} \quad & \Phi \\ \text{s.t.} \quad & p_k \leq p_{\max} \\ & \gamma_{\min} \leq \gamma_k \end{aligned} \quad (20)$$

where $\gamma_{\min}$ represent the minimum SINR boundary.

In this article, the transmit powers of all UEs are reduced continuously by alternating iterations until the EP of all UEs is equal to $\mathcal{E}_{\max}$. At this point, the SINR of all UEs is at the minimum value $\gamma_{\min}$, the transmit powers across all users are set to the minimum required to meet the URLLC reliability requirement, and a further reduction in the power of any UE would compromise the system's reliability. Therefore, we define $\mathbf{P}^* = [p_1^*, \dots, p_K^*]$ represents the optimal solution of (19), i.e., the optimal power allocation [47]. Specifically, we design a two-stage iterative optimization algorithm. The first stage ensures that all UEs meet URLLC, and the second stage maximizes EE under the URLLC-satisfied condition. The process of the algorithm is introduced as follows:

Initialize all parameters, including the SINR vector of all UEs $\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_K] \in \mathbb{R}^{1 \times K}$, the minimum SINR $\gamma_{\min}$, and the difference vector between current and the minimum SINR, defined as $\Delta \boldsymbol{\gamma} = [\Delta \gamma_1, \dots, \Delta \gamma_K] \in \mathbb{R}^{1 \times K}$, where $\Delta \gamma_k = \gamma_k - \gamma_{\min}$ ($\Delta \gamma_k \in \Delta \boldsymbol{\gamma}$). Setting δ as a constant exceedingly close to 0 and the transmit power of all UEs to $p_{\max}$.

Stage 1: Before the start of each iteration, two conditions are considered to address system reliability, which might

$$\begin{aligned} \gamma_k &= \frac{|\mathbb{E}\{DS_k\}|^2}{\sum_{i=1, i \neq k}^K \mathbb{E}\{|UI_{k,i}|^2\} + \sum_{i=1}^K \mathbb{E}\{|EM_{k,i}|^2\} + \mathbb{E}\{|AW_k|^2\}} \\ &= \frac{p_k \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}^2}{\sum_{i=1, i \neq k}^K p_i \sum_{m=1}^M \alpha_{m,i} \hat{\mu}_{m,i,k} + \frac{K}{\tau_{(p)}} \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k} + \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}}. \end{aligned} \quad (10)$$

have been overlooked in the initial power allocation. These conditions involve:

Condition 1: If $\forall \Delta\gamma_k \in \Delta\gamma, \Delta\gamma_k \geq 0$, all UEs already satisfy URLLC's reliability requirements. Therefore, the focus shifts to enhancing EE through the steps outlined in Stage 2.

Condition 2: If $\exists \Delta\gamma_k \in \Delta\gamma, \Delta\gamma_k < 0$, some UEs do not meet the reliability requirements. First, we compute $\Delta\gamma^1 = [\Delta\gamma_1, \dots, \Delta\gamma_U] \in \mathbb{C}^{1 \times U}$ and $\Delta\gamma^2 = [\Delta\gamma_1, \dots, \Delta\gamma_N] \in \mathbb{C}^{1 \times (K-U)}$, where $\Delta\gamma^1$ and $\Delta\gamma^2$ denote the proportion of UEs in $\Delta\gamma$ that satisfy and do not satisfy the reliability requirements, respectively. Two subcondition are discussed next:

Subcondition 2.1: If $\exists \Delta\gamma_{u^*} \in \Delta\gamma^1, \Delta\gamma_{u^*} \geq \delta, p_{u^*}$ is reduced to the optimal value by the binary search algorithm [48]. It is important to note that a reduction in p_{u^*} could potentially lead to an increase in $\Delta\gamma_l$, where $l \neq u^*$, as well as an increase in the length of $\Delta\gamma^1$. Consequently, this process allows more UEs to meet the requirements of URLLC.

Subcondition 2.2: If $\forall \Delta\gamma_{u^*} \in \Delta\gamma^1, 0 \leq \Delta\gamma_{u^*} < \delta$ or length of $\Delta\gamma^2$ is K : to ensure that existing UEs meeting reliability requirements are unaffected, we compute a new power sequence $\mathbf{P}'$, where $\forall p'_n \in \mathbf{P}', \Delta\gamma_n \in \Delta\gamma^2$ and $n \neq \text{argmax}(\Delta\gamma^2)$. A binary search algorithm is employed to simultaneously reduce $\mathbf{P}'$ until ensuring that $0 \leq \Delta\gamma_{\text{argmax}(\Delta\gamma^2)} < \delta$. This procedure optimizes not only $\Delta\gamma_{\text{argmax}(\Delta\gamma^2)}$, but also increases the value of elements in $\Delta\gamma^1$. The subsequent iteration prioritizes optimization for UEs in $\Delta\gamma^1$. Recalculate γ and $\Delta\gamma$ until satisfying *Condition 1*.

Stage 2: First, identify $g^* = \text{argmax}(\Delta\gamma)$. Subsequently, the binary search algorithm is utilized to reduce p_{g^*} until $0 < \Delta\gamma_{g^*} < \delta$. Then, recalculate γ and $\Delta\gamma$. Once the condition $\forall \Delta\gamma_k \in \Delta\gamma, \Delta\gamma_k \leq \delta$ is met, the iteration is terminated. At this point, all powers are optimized. Finally, return the current power allocation $\mathbf{P}^*$.

The pseudo-code for this procedure is presented in Algorithm 1.¹

IV. DEEP LEARNING FOR RESOURCES ALLOCATION

Conventional resource allocation methods typically employ alternating iterations to find the optimal strategy. However, these iterative processes involve numerous repeated computations, leading to an extensive resource overhead. Furthermore, the variable convergence time of iterative methods poses challenges for building a continuous and stable communication system [9], [10], [11].

To address these challenges, we apply supervised deep learning to find the optimal resource allocation strategy. Specifically, a RACNN is proposed to map channel states to

¹To satisfy the constraint in (20), condition $p_{\max} \geq (\gamma_{\min} I_k / C_k)$ must hold, where $C_k = |\sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}|^2$ represents the signal gain coefficient, $I_k = \sum_{i=1, i \neq k}^K P_i \sum_{m=1}^M \alpha_{m,k} \hat{\rho}_{m,i,k} + (K/\tau(p)) \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k} + \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}$ is the sum of all interference and noise. Furthermore, to prevent the algorithm from failing to converge when the problem is infeasible and to ensure that more UEs satisfy the constraints, the algorithm terminates and returns the power allocation when the iteration process exceeds 200 steps.

Algorithm 1 Two-Stage Power Allocation Algorithm

```

1: Initialize: Set  $p_k = p_{\max} = 20$  dBm,  $\delta = 0.00001$ .
   Compute  $\gamma, \gamma_{\min}, \Delta\gamma$ 
2: repeat
3:   if  $\forall \Delta\gamma_k \in \Delta\gamma, \Delta\gamma_k \geq 0$  then
4:      $g^* = \text{argmax}(\Delta\gamma)$ . Using binary search algorithm
       to reduce  $p_{g^*}$  until  $0 \leq \Delta\gamma_{g^*} < \delta$ 
5:   else
6:     Compute  $\Delta\gamma^1$  and  $\Delta\gamma^2$ 
7:     if  $\exists \Delta\gamma_{u^*} \in \Delta\gamma^1, \Delta\gamma_{u^*} \geq \delta$  then
8:        $g^* = \text{argmax}(\Delta\gamma^1)$ . Using binary search al-
         gorithm to reduce  $p_{g^*}$  until  $0 \leq \Delta\gamma_{g^*} < \delta$ 
9:     end if
10:    if  $\forall \Delta\gamma_{u^*} \in \Delta\gamma^1, 0 \leq \Delta\gamma_{u^*} < \delta$  or  $\text{length}(\Delta\gamma^2)$ 
        $= K$  then
11:       $l^* = \text{argmax}(\Delta\gamma^2)$ . Compute  $\mathbf{P}'$ , and using
        binary search algorithm to reduce  $\mathbf{P}'$  until
         $0 \leq \Delta\gamma_{l^*} < \delta$ 
12:    end if
13:  end if
14:  Compute  $\gamma$  and  $\Delta\gamma$ 
15: until  $\forall \Delta\gamma_k \in \Delta\gamma, \Delta\gamma_k \leq \delta$ 
16: Return Optimal power allocation sequence  $\mathbf{P}^*$ 

```

the optimal power allocation. As shown in Fig. 5, first, consider that the optimal power allocation is not determined only by the channel states of each UE individually. The interference among UEs also impacts it. Therefore, we stitch the channel states of all UEs into 2-D global features. Subsequently, we employ a CNN framework to extract the deep semantics of these features. Moreover, current deep learning-based resource allocation models typically concentrate on mapping channel states to allocation strategies without adequately considering the connection between these strategies and the optimization targets, potentially leading to overfitting [16], [17], [18]. Therefore, multitask learning [49] architecture is proposed to learn the connection between the optimal power allocation and the maximum EE, which imposes additional constraints and accelerates model convergence. Moreover, the model fitting is accelerated by adaptive loss weights [50]. Finally, we introduce the concept of deep transfer learning, which enables RACNN to adapt to new IIoT communication scenarios rapidly [40]. Compared with traditional iterative algorithms, RACNN exhibits a notably quicker decision-making capability and consistently produces stable power allocation strategies in real-time scenarios. Moreover, RACNN boasts robust environmental adaptability. Therefore, it is exceptionally well-suited to support URLLC.

A. Feature Selection and Label Design

High-quality features and accurate labels are crucial in supervised training, enabling the network to make faster and more precise predictions and inferences [51]. In CF-mMIMO systems, UE interference impacts the final power allocation strategy. Therefore, we combine the channel states of all UEs simultaneously in a single network forward propagation,

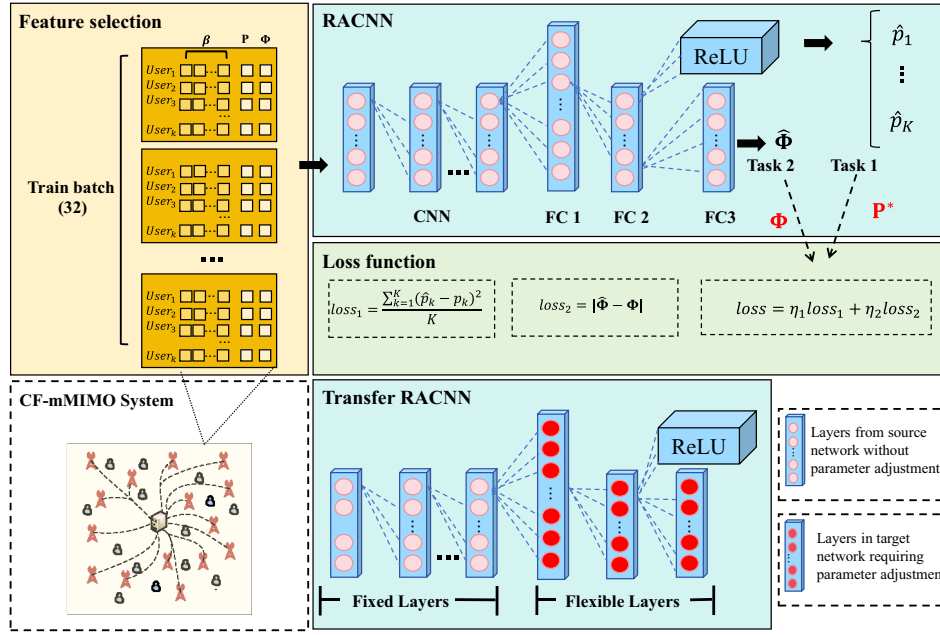


Fig. 5. RACNN network architecture.

TABLE I
STRUCTURAL PARAMETERS OF RACNN

Layer	Output
Input	$1 \times K \times M$
ResNet18 _{CNN}	$512 \times 4 \times 4$
Avgpool	$512 \times 1 \times 1$
FC1	256
FC2 + Relu	K
FC3	1

allowing RACNN to learn the correlation between different UEs. Extensive simulation data are obtained through Algorithm 1. As previously demonstrated, the large-scale fading coefficients $\beta_{m,k}$ correlate highly with the final power allocation strategy and can be predicted in advance by receiving terminals. As demonstrated in [38], we define the input features $\mathbf{F} \in \mathbb{R}^{K \times M}$ of a single forward transmission as

$$\mathbf{F} = \begin{bmatrix} \beta_{1,1} & \beta_{2,1} & \dots & \beta_{M,1} \\ \beta_{1,2} & \beta_{2,2} & \dots & \beta_{M,2} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{1,K} & \beta_{2,K} & \dots & \beta_{M,K} \end{bmatrix}. \quad (21)$$

The channel states of all UEs is contained in $\mathbf{F}$. The label of a single forward transmission is defined as $\mathbf{L} = [\mathbf{P}^*, \Phi]$, where $\mathbf{P}^*$ represents the optimal power allocation strategy, and Φ represents the maximum EE.

B. RACNN for Resources Allocation

Table I shows the structural parameters of RACNN. The global channel features are initially extracted using a CNN, which is capable of capturing deep semantic information from state space. Considering that traditional CNNs suffer from

vanishing or exploding gradients and have lower generalization performance. Referring to [52] and [53], we utilize the convolutional layers of the standard ResNet18 as the backbone network, which can improve the spatial information extraction through the synergistic application of the sliding convolution kernel and the residual architecture. Subsequently, the CNN output undergoes downsampling via FC1, and then the features are mapped to the approximately optimal powers $\hat{\mathbf{P}}^*$ by FC2. The mapping function is expressed as

$$\hat{\mathbf{P}}^* = \text{Relu}(\omega_2 \mathbf{X}_K^{\text{cnn}} + b_2) \quad (22)$$

where $\mathbf{X}_K^{\text{cnn}}$ represents the output vector of the CNN, ω_2 and b_2 represent the weight and bias of FC2, respectively, and $\text{Relu}(\cdot)$ is the activation function, aiming at preventing the predicted value from being less than 0.

Multitask training is a method whereby a neural network can simultaneously train to handle multiple related tasks by sharing specific parameters [45]. RACNN maps $\hat{\mathbf{P}}^*$ to approximate maximum EE through FC3. This process leverages increased data sources and training constraints to enhance the model's generalization ability, which can be expressed by

$$\hat{\Phi} = \omega_3 \hat{\mathbf{P}}^* + b_3 \quad (23)$$

where $\hat{\Phi}$ denotes the approximate maximum EE, ω_3 and b_3 represent the weights and bias of FC3, respectively.

The neural network obtains the approximate power allocation through a forward transmission process and then calculates the loss to quantify the discrepancy between the predicted value and the label [14]. The loss function of RACNN is expressed as

$$\text{loss} = \eta_1 \text{loss}_1 + \eta_2 \text{loss}_2 \quad (24)$$

where $\text{loss}_1 = [\sum_{k=1}^K (\hat{p}_k - p_k)^2 / K]$, $\text{loss}_2 = |\hat{\Phi} - \Phi|$, η_1 and η_2 represent the weights of each task loss, respectively.

This article employs the dynamic weight average method to achieve a more balanced training relationship between different tasks [54].

C. Deep Transfer Learning in Nonstationary Environments

Since RACNN is trained offline, its application is limited to specific CF-mMIMO systems. However, the communication environments of IIoT are highly dynamic and nonstationary, and many hidden parameters, such as small-scale fading coefficients and the number of antennas, are not included in the training features. If these parameters vary more than the robustness bound of the model, the model's fitting ability will inevitably decrease [40].

The most straightforward approach for accommodating time-varying communication scenarios is to train a new network from scratch. However, this approach requires a significant amount of training time and presents challenges regarding the availability of training data. To handle these problems, we propose a deep transfer framework to train a more effective RACNN with fewer samples [21].

The fundamental concept behind deep transfer learning is to transfer knowledge gained from a trained source task to a new target task [55]. Among the various methods, fine-tuning is widely recognized and utilized [37], [51]. A portion of the well-trained neural network parameters in the source task is reused in the target task neural network. Therefore, the requirement for the number of labeled training samples for the fine-tuned network is less than that for the network using random initialization. As shown in Fig. 5, we fixed the parameters of CNN layers, and all parameters of fully connected layers were updated throughout the new training process.

V. NUMERICAL RESULTS

This section presents numerical results to validate the performance achievable by the proposed schemes. Specifically, we define three cases, designated Case 1, Case 2, and Case 3, which represent three smart factory scenarios deploying CF-mMIMO systems with radii of 100, 250, and 500 m, respectively. In the system, M AP, equipped with L antennas and serve K single-antenna UEs. APs are uniformly deployed within the coverage range, with their positions fixed. In contrast, UEs are uniformly and randomly distributed across the region. First, We conduct comparative experiments on different numbers of UEs K , the blocklength of data $\tau_{(s)}$, the EP boundary $\mathcal{E}_{\max}$, and the maximum transmission powers $p_{\max}$ for each, verifying the effectiveness of the proposed optimization algorithm, where corresponding simulation results are presented in Figs. 6–9. Subsequently, we train a RACNN to approximate the optimal power allocation in varying numbers of UEs and then observe the difference between the output power of the RACNN and the optimal power. The relevant experimental results are shown in Figs. 10–12. In addition, we analyze the complexity of the proposed schemes. Finally, the RACNN model based on deep transfer learning

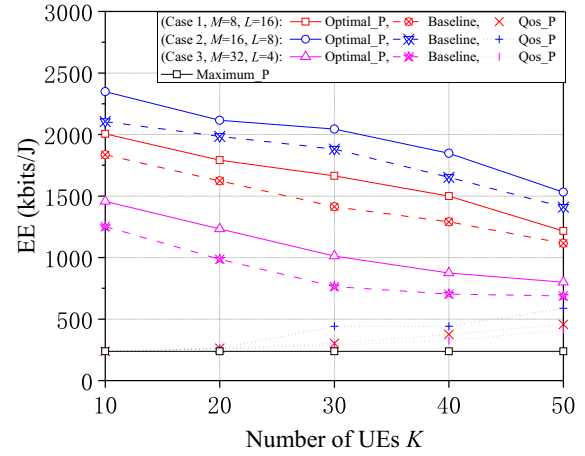


Fig. 6. Performance comparison of different power allocation schemes with different number of UEs. ($p_{\max} = 20$ dBm, $\tau_{(s)} = 3000$ CUs, $\mathcal{E}_{\max} = 10^{-5}$).

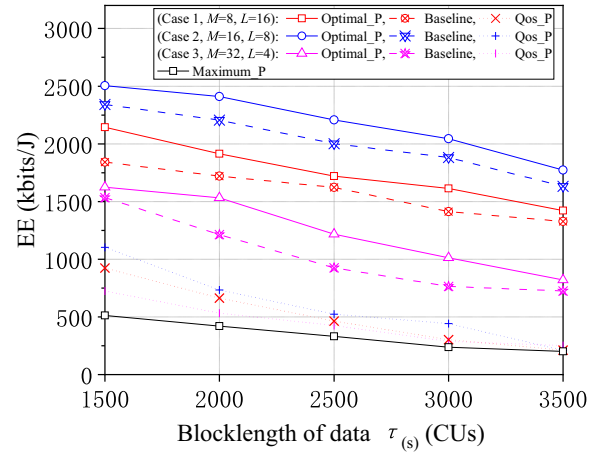


Fig. 7. Performance comparison of different power allocation schemes with different blocklength of data ($p_{\max} = 20$ dBm, $K = 30$, $\mathcal{E}_{\max} = 10^{-5}$).

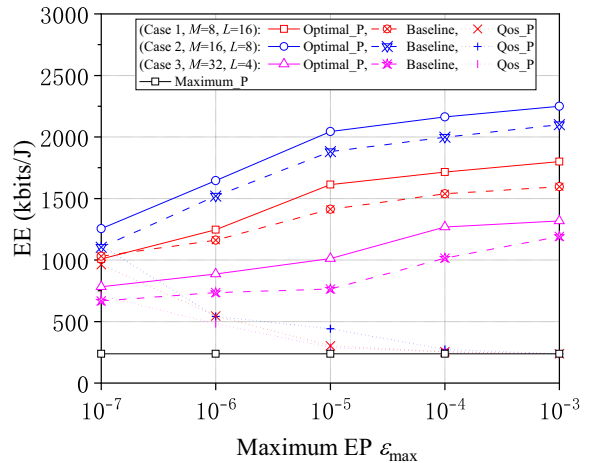


Fig. 8. Performance comparison of different power allocation schemes with different maximum EP ($p_{\max} = 20$ dBm, $\tau_{(s)} = 3000$ CUs, $K = 30$).

is conducted, and we verify its effectiveness under different antenna numbers L . The experimental results are depicted in Figs. 13 and 14.

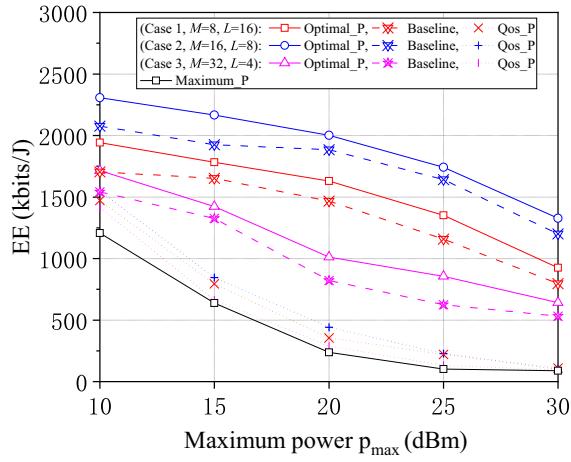


Fig. 9. Performance comparison of different power allocation schemes with different maximum power ($\tau_{(s)} = 3000$ CUs, $K = 30$, $\mathcal{E}_{\max} = 10^{-5}$).

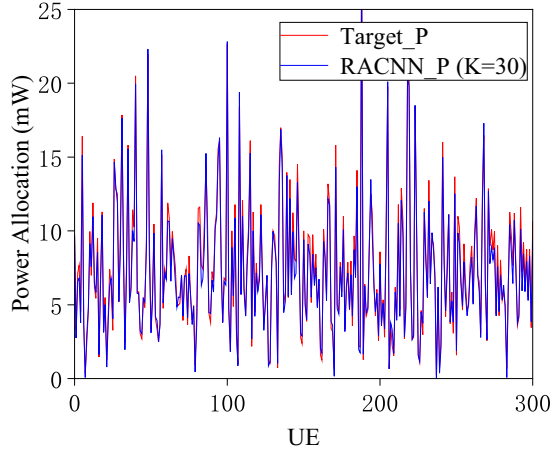


Fig. 10. Power distribution comparison ($K = 30$, $M = 32$, $\tau_{(s)} = 3000$ CUs, $L = 4$, $p_{\max} = 20$ dBm, $\mathcal{E}_{\max} = 10^{-5}$).

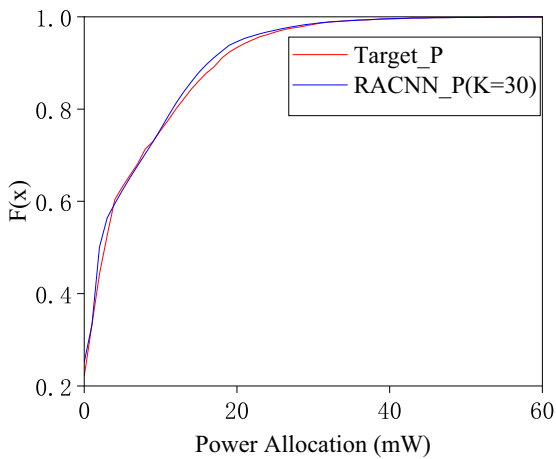


Fig. 11. CDF of power allocation ($K = 30$, $M = 32$, $\tau_{(s)} = 3000$ CUs, $L = 4$, $p_{\max} = 20$ dBm, $\mathcal{E}_{\max} = 10^{-5}$).

A. Details of Experiments

The experiments were conducted on a host computer with an Intel i7, 2.10-GHz CPU, 32.0-GB RAM, GTX4060 GPU,

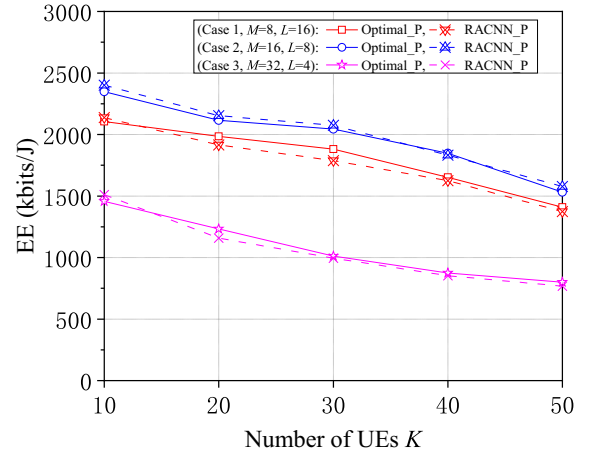


Fig. 12. Performance comparison of RACNN with different optimization strategies for different numbers of UEs ($\tau_{(s)} = 3000$ CUs, $p_{\max} = 20$ dBm, $\mathcal{E}_{\max} = 10^{-5}$).

TABLE II
SIMULATION PARAMETERS

Parameters	Value
Carrier frequency (GHz)	2.0
Bandwidth (MHz)	10
Frame length of data (bits)	32
Frame length of Pilot (bits)	64
End-to-end latency (ms)	0.5
Noise power spectral density (dBm/Hz)	-174

and Windows 11 (64 Bits) OS. PyCharm 2023.2.1 was used for the algorithm and deep learning experimental platform. The neural network was constructed using PyTorch 2.0.1, and the Adam optimizer was employed to accelerate convergence during training. The RACNN model was trained over 600 epochs, with a batch size of 32, an initial learning rate of 0.01, and a training-to-test set ratio of 8:2, the Rician factor ρ of 10, the total antenna numbers of 128. Detailed simulation parameters are provided in [56, Table II]. The experiments utilized the following large-scale model (in dB) [57]

$$\beta_{m,k} = -30.5 - 37.6 \log_{10}(d_{m,k}) + F_{m,k} \quad (25)$$

where $d_{m,k}$ represents the straight line distance (in m) between the k th UE to the m th AP, and $F_{m,k}$ denotes the shadow fading component, which follows a Gaussian distribution with $F_{m,k} \sim \mathcal{N}(0, 8)$.

B. Validation of the Optimal Power Allocation Algorithm

This article uses the EE optimization algorithm proposed in [44] as the baseline method, which use the golden section method to iterative alternate and updating the transmit power and the time-frequency block lengths for EE optimization under URLLC constraints, achieving at least a locally optimal solution. Additionally, maximum power allocation (Maximum_P) and URLLC-satisfied power allocation (Qos_P) are used for comparison. Qos_P represents the power allocation at the end of Stage 1 in Algorithm 1, ensuring that all UEs fully satisfy the QoS requirements for URLLC. It is important to note that if the initial power allocation already

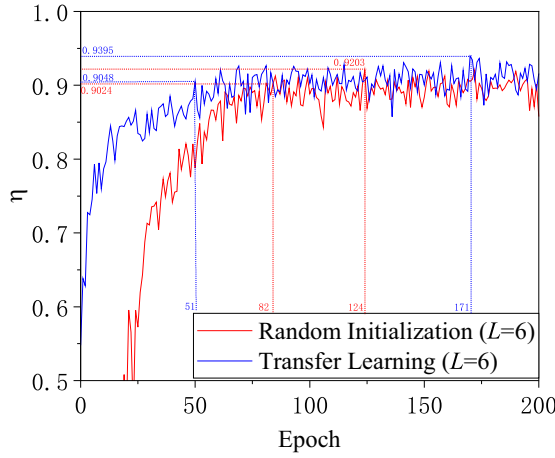


Fig. 13. Comparison of the deep transfer learning and random initialization parameter training processes ($L = 6$).

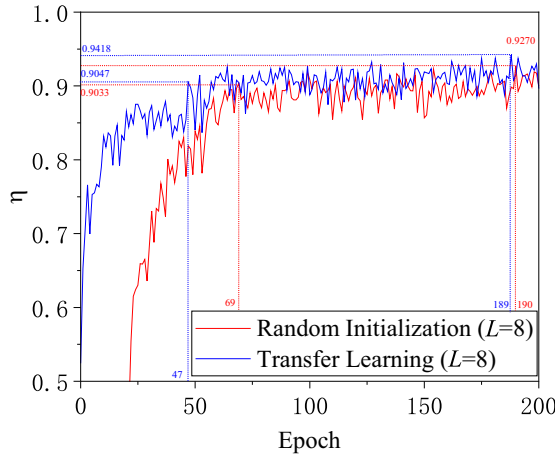


Fig. 14. Comparison of the deep transfer learning and random initialization parameter training processes ($L = 8$).

meets the URLLC requirements, Qos_P becomes equivalent to Maximum_P.

Fig. 6 compares the performance of different allocation schemes with different numbers of UEs K . We observe that the EE of Optimal_P gradually decreases as K increases. This trend arises because the increase in K leads to higher interference among UEs, thereby reducing signal quality and necessitating higher transmit power to compensate for signal losses. Furthermore, increased interference may cause more UEs to have initial power that cannot meet URLLC requirements. Therefore, the EE of Qos_P tends to increase with K . In Case 1 ($M = 8$, $K = 30$, $L = 16$), the EE of Optimal_P is 601.69%, 450.66%, and 17.69% higher than Maximum_P, Qos_P, and baseline method, respectively. In Case 2 ($M = 16$, $K = 30$, $L = 8$), it is 762.45%, 363.49%, and 8.6% higher, respectively.

With our proposed schemes, Fig. 7 reflects the improvement effects on varying blocklength of data $\tau_{(s)}$. From (10) and (15), an increase in $\tau_{(s)}$ leads to a decrease in data rate. Therefore, as $\tau_{(s)}$ increases, the EE of Optimal_P shows a declining trend. Moreover, the increase in $\tau_{(s)}$ may also result in higher channel estimation error, resulting in the reducing of SINR. Thus, the

trend of Qos_P is also decreasing. In Case 3 ($M = 32$, $L = 4$, $\tau_{(s)} = 2000$ CUs), the EE of Optimal_P is 263.9%, 187.43%, and 26.19% higher than Maximum_P, Qos_P, and baseline method, respectively.

For our proposed schemes, Fig. 8 compares the performance of different allocation schemes with different maximum EP $\mathcal{E}_{\max}$. According to (16), as $\mathcal{E}_{\max}$ increases, the minimum SINR decreases, resulting in a decrease in optimal power. Consequently, the EE of Optimal_P shows an upward trend. An increase in $\mathcal{E}_{\max}$ also means that the UE is more likely to guarantee the URLLC requirements, leading to a decrease in the EE of Qos_P. In Case 2 ($M = 16$, $L = 8$, $\mathcal{E}_{\max} = 10^{-4}$), the EE of Optimal_P is 812.66%, 689.41%, and 8.26% higher than Maximum_P, Qos_P, and baseline method, respectively.

Communication quality improves as the maximum power $p_{\max}$ increases, and UEs are more likely to meet URLLC requirements [5], [9]. Therefore, as observed in Fig. 9, Qos_P shows a decreasing trend. However, there is less room for powers to reduce due to the increasing interference among UEs, resulting in a declining trend in the EE of Optimal_P. The EE of Optimal_P is 179.03%, 124.56%, and 7.9% higher than Maximum_P, Qos_P, and baseline method in Case 1 ($M = 8$, $L = 16$, $p_{\max} = 15$ dBm), respectively.

C. Validation of RACNN for Optimal Power Allocation

In this subsection, we experimentally validated the effectiveness of the RACNN. Specifically, we construct the RACNN to approximate the optimal power allocation with different values of K and Cases ($\tau_{(s)} = 3000$ CUs, $p_{\max} = 20$ dBm, $\mathcal{E}_{\max} = 10^{-5}$). We developed a CF-mMIMO uplink system and derived the optimal power allocation strategy for training data generation. First, we generate the spatial locations of all APs and UEs in the system and calculate the channel coefficients according to (1) and (24), and then the optimal power allocation scheme is obtained by Algorithm 1. For each scenario, we collected 3000 simulation results, where each dataset includes channel information, optimal power allocation, and EE for 30000 to 150000 samples. These datasets were then used as inputs for neural networks and split into training and testing sets.

Fig. 10 shows the specific power allocation. We randomly selected ten instances from the test set at $K = 30$, including 300 samples. The result shows that RACNN aligns closely with the labels, with the mean square error (MSE) of 0.918 (mW^2). Furthermore, we compare the prediction and labeled powers' cumulative distribution functions (CDFs). Fig. 11 illustrates that RACNN accurately fits the optimal power across various scales. Fig. 12 compares the performance of RACNN's power allocation with the proposed optimal power allocation scheme at three cases. The results evident that RACNN's power allocation closely resembles EE, which is achieved by the optimal power allocation strategy. Moreover, the average EE differences is within 20 kbits/J.

D. Complexity Analysis

This section analyzes the complexity of the proposed optimal algorithm and RACNN. According to the iterative

TABLE III
FLOPS OF RACNN (M)

K	10	20	30	40	50
FLOPs	207.61	364.77	538.75	695.91	903.49

TABLE IV
COMPARISON OF DECISION RUNTIME (s)

K	10	20	30	40	50
RACNN _{GPU}	0.0007	0.0009	0.0018	0.0031	0.0067
RACNN _{CPU}	0.0029	0.0055	0.0092	0.0119	0.0152
Algorithm 1 _{CPU}	21.33	27.28	29.74	33.38	37.49
Baseline _{CPU}	23.52	32.11	35.72	41.29	45.01

TABLE V
COMPARISON OF TRAINING CONVERGENCE TIME (h)

K	10	20	30	40	50
Random Initialization	3.89	4.113	5.230	5.954	6.195
Transfer Learning	1.115	1.599	1.982	2.301	2.626

process outlined in Stage 1 of Algorithm 1, if some UEs initially fail to meet URLLC requirements, a binary search algorithm is employed to increase the number of UEs meeting these requirements. This process may involve multiple iterative loops with a complexity of $O(K - 1)$. Once all UEs satisfy URLLC requirements, Stage 2 commences with an iteration complexity of $O(K)$. Therefore, the overall complexity of our proposed algorithm is $O(K)$. Moreover, the baseline method has an approximate complexity of $O(K(N - K))$.

Floating-point operations per second (FLOPs) are precious for quantifying the computational demands of neural networks because these networks primarily perform matrix multiplications and convolution operations, which can be translated into counts of floating-point operations [58]. The FLOPs associated with RACNN across varying numbers of UEs K are detailed in Table III.

Furthermore, the computational speed of deep learning models can be significantly enhanced through the parallel processing capabilities of GPUs. It is worth noting that DNN typically has a large number of parameters, which often results in longer training times. However, in practical applications, the parameters of the RACNN are fixed, and the time cost of resource allocation strategies is determined by a single forward propagation, i.e., the decision-making time. Therefore, our primary focus is on comparing the decision-making speed of different schemes. We compared the average decision time between GPU-based RACNN, CPU-based RACNN, CPU-based proposed iterative algorithm, and CPU-based baseline method in the same experimental platform. As shown in Table IV, when K is 30, the running time of CPU-based RACNN is reduced by 99.97% and 99.98% compared with the proposed optimization algorithm and baseline algorithm [44], while GPU-based RACNN further shrinks by 80.43% compared to CPU. Consequently, employing DNN-based resource allocation proves more effective for power output.

E. Validation of Deep Transfer Learning

Consider that the channel state may occasionally change in real-world application scenarios. For example, specific APs may turn some antennas on or off. If the variation in optimal power is excessively large and exceeds the robustness range of the DNN, it could lead to performance degradation or system failure. Therefore, this section evaluates the performance of the proposed deep transfer learning method by varying the number of antennas L in the AP configuration.

Specifically, we use the RACNN trained in the previous section as the source network, and then a transfer RACNN is constructed as the target network for case 3 ($M = 32$, $\tau_{(s)} = 3000$ CUs, $p_{\max} = 20$ dBm, $\mathcal{E}_{\max} = 10^{-5}$). This experiment generated 800 scenario instances containing 24 000 samples with $L = 6$ and $L = 8$, respectively. To quantify the model's performance, the average approximation accuracy is defined as [21]

$$\eta = \frac{\sum_{k=1}^K 1 - \frac{|\hat{p}_k - p_k|}{p_k}}{K}. \quad (26)$$

Figs. 13 and 14 compare deep transfer learning with a benchmark model, where novel neural networks are trained using randomly initialized parameters (labeled as Random Initialization). Each parameter in the benchmark model is initialized with zero-mean and unit-variance Gaussian variables. From Fig. 13, we observe that deep transfer learning achieves an accuracy of approximately 0.9 after 51 epochs, while the random initialization method requires 82 epochs to achieve the same accuracy. Furthermore, the average accuracy of deep transfer learning is higher than the random initialization method during the training process. This result is because a large number of labeled samples (more than 70 000) were used to train the source network in the early stage, and the model is still robust even if the channel environment changes within a specific range. A similar conclusion is observed from the results of Fig. 14. Table V presents a comparison of convergence between the random initialization and transfer learning methods in a new environment at $L = 6$ (i.e., the training time required for the accuracy η to first reach 90%). The convergence time using the random initialization approach is approximately three times longer than that of transfer learning. Therefore, deep transfer learning can train a higher performance model with fewer samples in a shorter time than random initialization methods.

VI. CONCLUSION

This article presents an in-depth study to maximize EE under URLLC requirements in CF-mMIMO systems. We have first analyzed the relationship between EE, EP, and latency with short-packet transmission, and have designed a two-stage power-optimization algorithm to maximize the EE while guaranteeing URLLC demands. Then, we have proposed the DNN architecture (RACNN) to approximate the optimal allocation strategy for real-time and stable output. Finally, we have introduced the concept of deep transfer learning, which allows the model to adapt to changes in the communication environment. Furthermore, we have conducted

extensive simulation experiments to validate the efficacy of the proposed algorithm, NNs, and deep transfer learning. In the future, we will focus on the resource optimization problem in the same scenario but involving different QoS services, as well as the scalable inputs of the network. Furthermore, we will try to utilize the less complex algorithms to search for the optimal resource allocation.

APPENDIX

PROOF FOR THEOREM 1

To derive the closed-form expression for the SINR given in (10), we need to calculate $\mathbb{E}\{|DS_k|^2\}$, $\mathbb{E}\{|UI_{k,i}|^2\}$, $\mathbb{E}\{|EM_{k,i}|^2\}$, and $\mathbb{E}\{|AW_k|^2\}$. These calculations are performed under the assumption of a set of deterministic large-scale coefficients, with the statistics of the channel and estimation errors known at the terminals. For simplicity, let $\hat{\xi}_{m,k} = \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}|^2\}$, and $\hat{\mu}_{m,i,k} = \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,i}|^2\} (i \neq k)$.

1) *Compute $\mathbb{E}\{|DS_k|^2\}$:* From (1) and (4), we have $\mathbb{E}\{\hat{\mathbf{g}}_{m,k}\} = \mathbb{E}\{\mathbf{g}_{m,k}\} + \mathbb{E}\{\mathbf{n}_{m,k}\} = 0$, which leads to the following result:

$$\begin{aligned} \mathbb{E}\{|DS_k|^2\} &= \mathbb{E}\left\{\left|\sqrt{p_k} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,k}\right|^2\right\} \\ &= p_k \mathbb{E}\left\{\left|\sum_{m=1}^M \alpha_{m,k} \mathbb{E}\{\hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,k}\}\right|^2\right\} \\ &= p_k \mathbb{E}\left\{\sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}\right\}. \end{aligned} \quad (27)$$

2) *Compute $\mathbb{E}\{|UI_{k,i}|^2\}$:* Consider the property that if two random variables A and B are independent, with $\mathbb{E}\{A\} = 0$, then it follows that $\mathbb{E}\{|A+B|^2\} = \mathbb{E}\{|A|^2\} + \mathbb{E}\{|B|^2\}$. Since $\hat{\mathbf{g}}_{m,k}$ is independent of $\hat{\mathbf{g}}_{m,i}$, we have

$$\begin{aligned} \mathbb{E}\{|UI_{k,i}|^2\} &= \mathbb{E}\left\{\left|\sqrt{p_i} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,i}\right|^2\right\} \\ &= p_i \sum_{m=1}^M \alpha_{m,k} \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}^H \hat{\mathbf{g}}_{m,i}|^2\} \\ &= p_i \sum_{m=1}^M \alpha_{m,k} \hat{\mu}_{m,i,k}. \end{aligned} \quad (28)$$

3) *Compute $\mathbb{E}\{|EM_{k,i}|^2\}$:* As previously stated, $\mathbf{x}_{(p),i} \mathbf{x}_{(p),i}^H = |\mathbf{x}_{(p),i}^H|^2 = \tau_{(p)}$, and the estimation error $\mathbf{n}_{m,i}$ is given by $[\mathbf{W}_{(p),m} \mathbf{x}_{(p),i}^H \sqrt{p_i^{-1}} / \tau_{(p)}]$ with i.i.d. $\mathcal{CN}(0, [1/n p_k])$. Moreover, the pilot noise $\mathbf{W}_{(p),m}$ follows i.i.d. $\mathcal{CN}(0, 1)$ and $\mathbf{W}_{(p),m}$ is independent of $\hat{\mathbf{g}}_{m,k}$. Therefore, we have

$$\begin{aligned} \mathbb{E}\{|EM_{k,i}|^2\} &= \mathbb{E}\left\{\left|\sqrt{p_i} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \mathbf{n}_{m,i}\right|^2\right\} \\ &= \mathbb{E}\left\{\left|\sqrt{p_i} \sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \frac{\mathbf{W}_{(p),m} \mathbf{x}_{(p),i}^H \sqrt{p_i^{-1}}}{\tau_{(p)}}\right|^2\right\} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{\tau_{(p)}} \mathbb{E}\left\{\left|\sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \mathbf{W}_{(p),m}\right|^2\right\} \\ &= \frac{1}{\tau_{(p)}} \sum_{m=1}^M \alpha_{m,k} \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}^H \mathbf{W}_{(p),m}|^2\} \\ &= \frac{1}{\tau_{(p)}} \sum_{m=1}^M \alpha_{m,k} \mathbb{E}\{|\hat{\mathbf{g}}_{m,k}|^2\} \mathbb{E}\{|\mathbf{W}_{(p),m}|^2\} \\ &= \frac{1}{\tau_{(p)}} \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}. \end{aligned} \quad (29)$$

4) *Compute $\mathbb{E}\{|AW_k|^2\}$:* Similarly to (28), it can be proved as

$$\begin{aligned} \mathbb{E}\{|AW_k|^2\} &= \mathbb{E}\left\{\left|\sum_{m=1}^M \alpha_{m,k} \hat{\mathbf{g}}_{m,k}^H \mathbf{W}_{(s),m}\right|^2\right\} \\ &= \sum_{m=1}^M \alpha_{m,k} \hat{\xi}_{m,k}. \end{aligned} \quad (30)$$

From (27), (28), (29), and (30), we obtain (10). ■

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